

A COMMERCIAL ROVER FOR THE LUNAR VULKAN IMAGING AND SPECTROSCOPY EXPLORER MISSION TO GRUITHUISEN DOMES. J. D. Lawrence¹, J. Bailey¹, J. Cerny¹, Z. Fitzgerald¹, K. Klein¹, P. Ng¹, D. Sabahi¹, A. Shmavonian¹, N. Steele¹, B. Vogel¹, V. Vendiola¹, A. Yaghi¹, K. Zacny¹. ¹Honeybee Robotics Exploration Systems (jdlawrence@honeybeerobotics.com).

Introduction: Firefly Aerospace is flying six investigations to Gruithuisen Gamma Dome (36°N, 40°W) in 2028 for NASA's Commercial Lunar Payload Services (CLPS) CP-21 mission. The 97 kg science payload includes the Lunar Vulkan Imaging and Spectroscopy Explorer (Lunar-VISE) instrument suite. The Lunar-VISE investigation consists of two instruments (17.5 kg total) mounted to Firefly's lander and three instruments (up to 16 kg total) mounted to a rover, the first commercial rover contracted by NASA through the CLPS initiative [1].

Firefly Aerospace has engaged Honeybee Robotics, a Blue Origin Company, to provide the rover. Here we describe the mobile concept of operations and architecture in support of the Lunar-VISE investigation.

The CP-21 Rover Mission: The Gruithuisen Domes on the Moon are thought to be silica-rich volcanic features – but similar domes on Earth only form through processes driven by interactions between plate tectonics and liquid water. The Lunar-VISE investigation aims to resolve the mystery of how the lunar domes could have formed in the absence of plate processes and volatiles.

The rover will carry the Lunar-VISE (LV) VNIR Imaging Camera (LV-VIC), a multispectral camera and radiometer called Compact Infrared Imaging System (LV-CIRiS), and the Gamma Ray and Neutron Spectrometer (LV-GRNS). The rover will complete an up to 1.1 km traverse over one lunar day and operate at a radius of up to 500 m from Firefly's lander. The onboard Lunar-VISE instruments will collect high-resolution multispectral imagery and map the morphology, composition, and abundance of surface and near-surface minerals and elements along the traverse [2].

The Lunar-VISE team will also characterize spatial variability in geotechnical regolith properties along the traverse [2]. Both the rover and the lander-mounted Lunar-VISE context camera will provide high-resolution observations of interactions between the rover wheels and lunar regolith, in the context of paired rover and drive system telemetry (i.e., power or torque at each wheel).

The CP-21 Rover: The rover is a four wheel, solar-powered vehicle with a rechargeable battery. It is designed to cover multiple km during the lunar day and can traverse 20° slopes. The rover uses skid steering

with a dedicated drive actuator for each wheel to enable forward, reverse, and turn in place operation.

Firefly's lander provides a communications relay for science data and rover command and control. In addition to the Lunar-VISE instruments, the rover will carry high resolution cameras, lights, and navigation sensors for localization relative to the Firefly lander.



Figure 1. The Firefly lander and Honeybee Robotics rover for CP-21 and the Lunar-VISE investigation into the geologic origins of the Gruithuisen Domes. Image: [Firefly Aerospace](#).

References: [1] NASA (2024) [nasa.gov/news-release/more-nasa-science-tech-will-fly-to-moon-aboard-future-firefly-flight/](https://www.nasa.gov/news-release/more-nasa-science-tech-will-fly-to-moon-aboard-future-firefly-flight/). [2] Donaldson Hanna K. et al. (2023) *LPSC LIV*, 2806.